Statistics 4365/6365 – Modern Statistical Programming - Fall 2025

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Office Hours:
Monday 1-2 by Zoom: https://zoom.us/j/98658152868
Thursday after class in person
By appointment.

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

TA: Yumiao Lei
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Office Hours:
In-person: Friday, 2:00–3:00 PM (Office: Room 517)
Online (Zoom): Wednesday, 2:00–3:00 PM
Zoom link:
https://zoom.us/j/92418667713?pwd=SMBN0n8DKTn6nA1jXHXHFjrZ3oZ85g.1

Textbook: We will use two books that are freely available online:

1) R for Data Science by Hadley Wickham and Garrett Grolemund. The full text is available at https://r4ds.had.co.nz/.
2) Data Science Using Python and R by Chantal Larose and Daniel Larose. It is available via the UGA library. Go to the library website, search the title, and you can download it. You may need to be on-campus for it to work.

Prerequisite: The undergraduate requirement is STAT 2360. There is no formal requirement for grad students. I will assume that you have a basic understanding of R. This means that you know how to write simple R scripts, use functions, and have some familiarity with R variable types like vector, data frame, etc. If you have not used R but have experience in some programming language other than R, then you will probably be fine. If you have no programming experience, this may not be the right course for you. I do not assume any prior knowledge of Python. I also assume that you have some familiarity with basic statistical procedures. If in doubt, talk to me.

Student Learning Outcomes

- Students will read in data, clean it, and format it for analysis in R and Python.
- Students will implement complex data visualizations in R and Python.
- Students will manipulate data sets to extract the information needed for a desired analysis in R and Python.

- Students will use standard programming techniques such as loops, conditional execution, and functions in complex data analysis in R and Python.
- Students will create user-friendly graphical interfaces to programs conducting complex data analysis and visualization.
- Students will identify and correct problems in their computer programs.
- Students will access a data base, run queries, and write simple programs using SQL.

Grading:
Homework: 60%
In-Class Exercises using Top Hat: 5%
Midterm exam: 10 %
Final Project: 25% of Grade

STAT 4365 versus 6365
Assignments and content will be the same between the undergraduate and graduate versions of the course. However, grading standards will be higher for students enrolled in the graduate version than for students enrolled in the undergraduate versions.

Top Hat and in-class exercises
This course will use Top Hat (www.tophat.com) for in-class exercises. Top Hat is required if you want to receive credit for in-class exercises, which are 5% of the grade.

To sign up, go to the Course Administration folder on eLC, then to Top Hat Setup, and then click the link. This should enroll you in the Tophat course.

There will be no makeup for missed in-class exercises. However, I will drop the lowest 20% of scores for each student to allow for missed classes or other issues.

Homework
There will be homework assignments every 1-2 weeks. These will typically involve applying programming techniques that we learn in class to data sets. Many assignments will involve exploring data sets and writing short reports explaining what you have found.

Note: The purpose of this class is to learn to manipulate data using computer programs. This means that you cannot in any way do data manipulations "by hand". You cannot do any calculations outside of your program. You cannot make any changes to data sets. Your code must always use the original data set. Everything must happen inside of your code.

Late Policy: Unless you make prior arrangements, there will be a 10% late penalty if assignments are not turned in by midnight of the day that they are due. There will be an additional 10% for each additional 24 hours that they are late (that is, the penalty is

incurred at midnight each day). You must turn in all required files in the correct format by the deadline or the late penalty will be applied. It is your responsibility to make sure that everything is correctly submitted. If there is a valid reason (illness, travel, etc) that you will not be able to turn in the assignment on time, email me about an extension. Extensions **must be approved ahead of a time** and must be requested no later than 24 hours in advance.

Warning #1: The homework assignments are challenging and you will need the full time. You may often need to come to me or the TA to ask questions about how to do things. You are doomed if you wait until the last minute to start the assignments.

Warning #2: There will be around 6-8 homework assignments that together comprise 60% of your grade. For example if there are 8 assignments, then each will be worth more than 7% of your grade. This means that missing a single assignment will drop your final grade by close to a full letter grade (e.g. A to B+ or B). **Do not miss even a single assignments if you want a good grade.**

As a University of Georgia student, you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty," and the Student Honor Code. All academic work must meet the standards described in "A Culture of Honesty" found at: www.uga.edu/honesty. Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

Midterm Exam
There will be an in-class midterm exam on Thursday October 16. The purpose of the exam will be to test your ability to program without the aid of the internet, AI, or other people. The exam will be designed to be easy for students who have been doing their own work in the course. You will not be allowed any use of the internet or any other outside resources.

Final Project:
The final project will synthesize techniques learned over the course of the semester in analyzing complex data sets. You will make a web app that does data visualization of some complex data set. I will give you more information about the final project later in the semester. **The final project will be due on Thursday December 11 by noon.**

eLearning Commons: I will use eLearning Commons for distribution of course material and for course communications. Information about your grade will also be posted here. The website is https://www.elc.uga.edu. All UGA students have an eLC account and you should see this course when you log in.

Software Installation
You should install the following software on your computer:

For Python, we will use Anaconda. You can download this at

Free Download | Anaconda
Just download, double click, and follow the instructions,

For R, we will use RStudio. You can download this at
https://www.rstudio.com/products/rstudio/download/
Again, just download, double click, and follow the instructions.
Let me know if you have any questions or problems.

Outside Resources and AI

Everyone who codes uses the internet to figure out how to do things. Knowing how to look things up is a necessary skill for a programmer. However, there is a difference between using the internet as an occasional aid to programming and using it as a substitute for thinking. Too many students spend hours googling for a solution rather than figuring out how to do it themselves. Coding is something that you learn by doing, not by googling. Spend your time trying things for yourself. If it doesn't work, figure out why it doesn't work and then fix it. A large part of learning to code (and doing mathematical things in general) is learning the mental skills and the mental fortitude to keep hammering at a problem until you solve it. If you solve the problem yourself, you will understand it and you will know what to do when you encounter it in the future. If you find the solution on the internet, you probably won't understand it and you also won't be able to tell whether it is actually correct.

This issue is magnified by ChatGPT and other large language models (LLMs). ChatGPT can write code and do many other things for you. Sometimes it does brilliantly. However, it also gets it completely wrong on a regular basis. Consider this case (from law, not programming):
ChatGPT: US lawyer admits using AI for case research - BBC News

A lawyer representing a client suing an airline submitted a legal brief that was written by ChatGPT. Unfortunately for the lawyer, several of the cases cited in the brief were fake. ChaptGPT had simply made them up and the lawyer is facing grave professional consequences as a result.

Or consider this one:
Chatbots Can Go Into a Delusional Spiral. Here's How It Happens. - The New York Times

ChatGPT convinced this poor guy that he had discovered a novel mathematical framework that was going to change the world and make him rich. Suffice it say that it was all fantasy.

AI tools are good for generating ideas, but absolutely cannot be trusted to be accurate. I have experimented with using ChatGPT with coding problems similar to what we do in this class. It produces code that is wrong a substantial percentage of the time. A lot of people in the programming world have been experimenting with ChatGPT and the consensus seems to be that it usually does well with simple things, but often makes mistakes with more complex things. Even with the simple things it makes mistakes often enough that you cannot depend on it.

Furthermore, it is not easy to tell that is wrong. It produces nice looking code that runs and produces output that often seems reasonable. In some cases I was only able to tell that it was wrong because I had working code to compare it to. The bottom line is that

depending on ChatGPT to write code is a bad idea: it will give you bad results much of the time.

The more important reason to not use ChatGPT for this class is because if you do you won't learn programming skills. We are at the beginning of a major shift in how people work in many fields and no one knows what the landscape will be in two years, let alone ten or twenty. There is no doubt that LLMs will become an important part of the programmer's toolkit and you should be learning to use them. However, you need to learn to think and solve problems on your own. If all that you can do is type things into ChatGPT, then what value do you add? Why would anyone hire you? If you want to be valuable in the AI world, then AI needs to be part of your toolkit but not a substitute for your own brain. The only way that you can learn to use your brain effectively in the context of programming is to learn to program. The only way to do that is to spend a lot of time figuring out how to program on your own.

AI Policy:

I want you to learn to use AI as a tool, but not as a substitute for your own brain.

- It is okay to use AI to answer focused questions about specific programming tasks.
- Is NOT okay to use AI to write your code or tell you how to do a problem.
    - You should never turn in code generated by AI.
    - You should never plug problems from this class into an AI.
    - You should never use AI programming assistants in this class.
    - If you use AI in an assignment, you are required to document. I will give detailed instructions in the first assignment.

You may not represent output generated by an AI tool as your own work. Any such use of AI output must be appropriately cited or disclosed, including quotation marks and in-line citations for direct quotes. Including anything you did not write in your assignment without proper citation will be treated as an academic misconduct case. Suspected unauthorized assistance, plagiarism, or other violations of UGA's "A Culture of Honesty," will be reported to the Office of Academic Honesty. For full details on how to properly cite AI-generated work, please see the APA Style article, How to Cite ChatGPT.

If you are unsure where the line is between collaborating with AI and copying from AI, I recommend that you do not have your assignment and the AI tool open on your device at the same time. Instead, take notes in your own words while you interact with the AI tool, then use your notes to remind you of what you've learned and to inform your work. Never copy output from AI tools into your assignment. Instead, use your interaction with the tool as a learning experience, then close the interaction down, open your assignment, and let your assignment reflect your improved understanding.

Finally, AI is highly vulnerable to inaccuracy and bias. You should assume AI output is wrong unless you either know the answer or can verify it with another source. It is your responsibility to assess the validity and applicability of any AI output used.